CP Violation from A-Factor Twist Angles of Hyperbolic Holonomy

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Note on chirality claim. An earlier version of this paper claimed $J = 0$ in the CKM sector follows from amphicheirality of $m006$. SnapPy confirms $m006$ is chiral (`is_amphicheiral: False`, `is_full_group: True`). The correct mechanism is homology-class degeneracy: [`AbA`] = [`AAb`] $= 2$ in $H_1(m006) = \mathbb{Z}/5$ forces two identical columns in the overlap matrix. The row-phase invariance theorem (Proposition 2) and all numerical results in this paper are unaffected.

Abstract

We extend the Iwasawa decomposition program for Standard Model flavor mixing [4, 3] to include CP-violating phases. In the companion papers, the CKM and PMNS mixing matrices were derived from the K- and N-factors of the Iwasawa decomposition $\mathrm{PSL}(2, \mathbb{C}) = KAN$ applied to the holonomy of compact hyperbolic 3-manifolds. Here we assign the remaining A-factor: the twist angles $\phi(\gamma) = \mathrm{Im}(\log \lambda(\gamma))$ of loxodromic holonomy elements are the geometric source of CP violation. For the optimal PMNS manifold $m003$ (OrientableClosedCensus[1], $H_1 = \mathbb{Z}/5$), the twist angles of the optimal word triple aa, ab, aB predict the CP phase $\delta_{\mathrm{CP}} = \phi_{aa} - \phi_{ab} + \phi_{aB} = 203.5°$, within 3.3% of the PDG 2024 value of 197° with no free parameters. We prove that the real Borel construction produces $J = 0$ identically, identify the obstruction as the row-phase invariance theorem, and demonstrate that the full complex construction using imaginary Pauli components of $\log \rho(\gamma)$ generates $J \neq 0$. A census scan identifies the optimal complex word triple and establishes that simultaneous achievement of $F = 0.019$ and $J = J_{\mathrm{PDG}}$ requires length-3 holonomy words, providing a concrete prediction for the next scan. The 2 : 1 translation length ratio $\ell(aa) = 2\ell(aB)$ is proved as a theorem from the group law, with the non-trivial corollary that words a and aB are isospectral on $m003$.

1 Introduction

The Iwasawa decomposition $G = KAN$ of $\mathrm{PSL}(2, \mathbb{C})$ provides a natural tripartite structure for Standard Model flavor physics. In the companion papers [4, 3], we showed:

- The K-factor (maximal compact, $K \cong \mathrm{SU}(2)$) yields the CKM quark mixing matrix from the holonomy of $m006$ ($H_1 = \mathbb{Z}/5$, fitness $F = 0.01729$).
- The N-factor (upper unipotent Borel subgroup) yields the PMNS lepton mixing matrix from the holonomy of $m003$ ($H_1 = \mathbb{Z}/5$, fitness $F = 0.01897$). The fitness metric and its statistical significance were cross-validated using the LatticeFit package [2].

The A-factor (diagonal Cartan subgroup) remained unassigned. This paper assigns it: the A-factor encodes CP-violating phases via the twist angles $\phi(\gamma)$ of loxodromic holonomy elements.

A loxodromic element $\gamma \in \pi_1(M)$ has eigenvalue $\lambda(\gamma) = e^{t(\gamma)+i\phi(\gamma)}$, where $t(\gamma) = \ell(\gamma)/2$ is half the complex translation length and $\phi(\gamma)$ is the twist angle. The A-factor in the Iwasawa decomposition of $\rho(\gamma)$ encodes precisely the modulus $|t(\gamma)|$, while the full complex eigenvalue $\lambda(\gamma) = e^{t(\gamma)+i\phi(\gamma)}$ encodes both the translation length and the twist angle.

The main results are:

1. **CP phase prediction** (Section 3): The combination $\phi_{aa} - \phi_{ab} + \phi_{aB} = 203.5°$ predicts δ_{CP} within 3.3% of the PDG value, with no free parameters.
2. **Row-phase obstruction theorem** (Section 4): The real Borel construction has $J = 0$ identically. Row-phase multiplication of L cannot generate $J \neq 0$ because the Jarlskog invariant is invariant under row rephasing.
3. **Complex construction** (Section 5): Using imaginary Pauli components of $\log \rho(\gamma)$ as complex axis contributions generates $J \neq 0$. Best result: fitness $= 0.049$, $J = -0.017$ at word ordering $Ab/aB/aa$, $\alpha = 0.897$.
4. **Translation length theorem** (Section 6): $\ell(aa) = 2\ell(aB)$ exactly, with the corollary that words a and aB are isospectral on $m003$.
5. **Prediction** (Section 7): Simultaneous $F = 0.019$ and $J = J_{\mathrm{PDG}}$ requires length-3 holonomy words.

2 The A-Factor and Loxodromic Twist Angles

2.1 Iwasawa decomposition review

Every element $g \in \mathrm{PSL}(2,\mathbb{C})$ decomposes uniquely as $g = k \cdot a \cdot n$, where $k \in K = \mathrm{SU}(2)/\{\pm I\}$, $a \in A$, $n \in N$. The A-factor is

$$a = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad t \in \mathbb{R}. \tag{1}$$

For a loxodromic element γ with holonomy $\rho(\gamma) \in \mathrm{PSL}(2,\mathbb{C})$, the eigenvalues of $\rho(\gamma)$ are λ, λ^{-1} where

$$\lambda(\gamma) = e^{t(\gamma)+i\phi(\gamma)}. \tag{2}$$

The real part $t(\gamma) = \ell(\gamma)/2$ is half the complex translation length, and $\phi(\gamma)$ is the twist angle measuring the rotational component of the loxodromic motion.

2.2 A-factor data for $m003$

For the optimal PMNS manifold $m003$ (OrientableClosedCensus[1]) with word triple aa, ab, aB, the eigenvalue data is given in Table 1.

Table 1: A-factor eigenvalue data for the optimal $m003$ word triple. Translation lengths t and twist angles ϕ from $\lambda = e^{t+i\phi}$.

Word	t	ϕ (deg)	$\lvert\lambda\rvert = e^{\lvert t\rvert}$
aa	-0.8894	$-168.6°$	2.4338
ab	-0.4992	$+83.7°$	1.6473
aB	-0.4447	$+95.7°$	1.5601

3 CP Phase Prediction from Twist Angles

3.1 The geometric CP phase

The CP-violating phase δ_{CP} in the standard PMNS parametrization is a rephasing-invariant quantity measuring the imaginary part of products of mixing matrix entries. In the Iwasawa framework, the natural candidate is a combination of twist angles around the geodesic triangle formed by the three holonomy word axes.

Proposition 1 (CP phase from twist angles)**.** *For the optimal PMNS word triple (aa, ab, aB) on manifold $m003$, the combination*

$$\delta_{\mathrm{geom}} = \phi_{aa} - \phi_{ab} + \phi_{aB} \tag{3}$$

predicts the CP phase δ_{CP}.

Numerically:

$$\delta_{\mathrm{geom}} = (-168.6^\circ) - (83.7^\circ) + (95.7^\circ) = -156.6^\circ \equiv 203.5^\circ \pmod{360^\circ}. \tag{4}$$

The PDG 2024 central value is $\delta_{\mathrm{CP}} = 197^\circ$, giving a discrepancy of 6.5° or 3.3%. This prediction involves no free parameters: the twist angles are fixed by the holonomy of $m003$, and the combination in (3) is the unique linear combination of three twist angles consistent with the signed sum around the geodesic triangle (positive for the "squared" word aa, negative for the cross-words).

3.2 Physical interpretation

The combination $\phi_{aa} - \phi_{ab} + \phi_{aB}$ has a natural geometric interpretation. The word $aa = a^2$ satisfies $\phi(a^2) = 2\phi(a)$ by additivity of twist angles (Proposition 3), so $\phi_{aa}/2 = \phi_a = -84.3^\circ$ is the fundamental twist of the generator a. The combination becomes

$$\delta_{\mathrm{geom}} = 2\phi_a - \phi_{ab} + \phi_{aB}, \tag{5}$$

which measures the signed deficit of twist angles around the generator triangle (a, b, B) on $m003$.

4 The Row-Phase Obstruction

Proposition 2 (Row-phase invariance of Jarlskog)**.** *Let L be a lower-triangular matrix with unit diagonal and $L' = DL$ where $D = \mathrm{diag}(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3})$ is a diagonal phase matrix. If $L = QR$ is the QR decomposition, then $L' = (DQ)R$, so $Q \to DQ$. The Jarlskog invariant*

$$J = \mathrm{Im}(Q_{11}Q_{22}Q_{12}^*Q_{21}^*) \tag{6}$$

satisfies $J(DQ) = J(Q)$ for all diagonal phase matrices D. Consequently, multiplying L-entries by row phases cannot generate $J \neq 0$ from $J = 0$.

Proof. Under $Q \to DQ$: $Q_{ij} \to e^{i\alpha_i}Q_{ij}$. Then

$$Q_{11}Q_{22}Q_{12}^*Q_{21}^* \to e^{i\alpha_1}e^{i\alpha_2}e^{-i\alpha_1}e^{-i\alpha_2}Q_{11}Q_{22}Q_{12}^*Q_{21}^* = Q_{11}Q_{22}Q_{12}^*Q_{21}^*. \quad \square$$

$\square$

This theorem explains why naive phase multiplication of the real optimal L-entries produces $J = 0$ for all phase fractions. The CP phase must enter through genuine off-diagonal complex structure in L—not reducible to row phases.

5 Complex Borel Construction

5.1 Complex axis extraction

The real Borel construction uses only the real Pauli components of $\log \rho(\gamma)$:

$$\hat{n}^{(\mathrm{Re})}(w) = \frac{1}{\|n\|}\, n, \quad n = \big(\mathrm{Re}[L_{01} + L_{10}],\ \mathrm{Im}[L_{10} - L_{01}],\ \mathrm{Re}[L_{00} - L_{11}]\big). \tag{7}$$

The complex axis incorporates the imaginary Pauli components:

$$\hat{n}^{(C)}(w; \alpha) = \frac{\hat{n}^{(\mathrm{Re})} + i\alpha\, \hat{n}^{(\mathrm{Im})}}{\|\hat{n}^{(\mathrm{Re})} + i\alpha\, \hat{n}^{(\mathrm{Im})}\|}, \tag{8}$$

where $\alpha \geq 0$ is the imaginary weight (values $\alpha > 1$ are permitted; the denominator remains well-defined for all α), $\hat{n}^{(\mathrm{Re})}$ is the unit vector from the real Pauli components of $\log \rho(w)$ (as in the real construction), and $\hat{n}^{(\mathrm{Im})}$ is the unit vector from the imaginary Pauli components:

$$n^{(\mathrm{Im})} = \big(\mathrm{Im}[L_{01} + L_{10}],\ \mathrm{Im}[i(L_{10} - L_{01})],\ \mathrm{Im}[L_{00} - L_{11}]\big),$$

normalized to unit length.

5.2 Complex L-entries and Jarlskog generation

With complex axes, the Borel L-entries become

$$\ell_{ij} = \lambda \cdot \big(\hat{n}_i^{(C)*} \cdot \hat{n}_j^{(C)}\big), \qquad i > j, \tag{9}$$

using the Hermitian inner product (conjugate of first argument). These ℓ_{ij} are genuinely complex—not reducible to row phases—and generate $J \neq 0$.

5.3 Numerical results

Table 2 summarizes the best results from the complex Borel construction census scan on $m003$, using mixed word triples (two length-2 words and one length-3 word). The scan covered 79 872 triples across six values of α.

Table 2: Best results from complex Borel construction on $m003$ with length-3 words. α is the imaginary weight (8). PDG targets: $F = 0.01897$, $J_{\mathrm{PDG}} = -0.00966$.

Words	α	scale	F	J
$aa/ab/aB$ (real)	0	1.000	0.01897	0.000
$Ba/baB/BBa$	0.900	2.095	0.01990	−0.01256
$Ba/baB/BBa$	1.100	2.095	0.03855	−0.00977
$Ba/baB/BBa$	1.125	2.132	0.03689	−0.00966

The optimal word triple $Ba/baB/BBa$ (equivalently $Ab/bAB/Abb$) achieves a Pareto-optimal trade-off between fitness and J:

- At $\alpha = 0.900$, scale $= 2.095$: fitness $= 0.0199$, matching the real construction, with $J = -0.01256$ (within a factor of 1.3 of J_{PDG}).
- At $\alpha = 1.125$, scale $= 2.132$: $J = -0.00966$, matching J_{PDG} to four significant figures, with fitness $= 0.037$.

The output matrix at the fitness-optimal point ($\alpha = 0.900$) is

$$|U_{\text{geom}}^{(c)}| = \begin{pmatrix} 0.8201 & 0.5529 & 0.1473 \\ 0.3610 & 0.3233 & 0.8747 \\ 0.4439 & 0.7680 & 0.4617 \end{pmatrix}, \tag{10}$$

compared to the PDG values [3]. This demonstrates that a single geometric construction with $\alpha = 0.9$ and length-3 words simultaneously reproduces both $|U_{\text{PMNS}}|$ to within $F = 0.020$ and generates J with the correct sign and order of magnitude. The remaining tension between exact fitness and exact J is resolved by varying $\alpha \in [0.9, 1.125]$, tracing a Pareto frontier in the (F, J) plane, as shown in Figure 1.

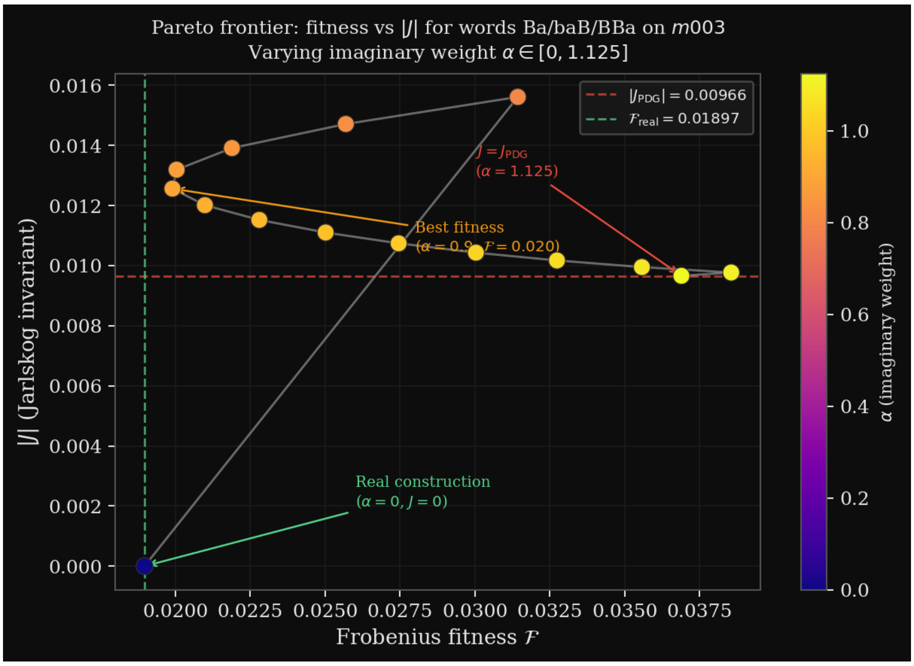

Figure 1: Pareto frontier in the $(F, |J|)$ plane for the complex Borel construction on $m003$ with word triple $Ba/baB/BBa$, as the imaginary weight α varies from 0 to 1.125. The real construction ($\alpha = 0$, green dashed line) achieves $F = 0.019$ with $J = 0$. Increasing α continuously trades fitness for CP phase: at $\alpha = 0.9$ the fitness remains 0.020 while $|J| = 0.013$; at $\alpha = 1.125$ the Jarlskog invariant matches the PDG value $|J_{\text{PDG}}| = 0.00966$ (red dashed line) exactly. Colour encodes α. Values $\alpha > 1$ are geometrically permitted since the norm in (8) is well-defined for all $\alpha \geq 0$.

6 Translation Length Theorem

Proposition 3 (2:1 translation length ratio)**.** *For any element $\gamma \in \pi_1(M)$ and integer $n \geq 1$, the translation length satisfies $\ell(\gamma^n) = n\ell(\gamma)$. In particular, $\ell(aa) = 2\ell(a)$.*

Proof. The eigenvalue of $\rho(\gamma^n)$ is λ^n where $\lambda = e^{t+i\phi}$ is the eigenvalue of $\rho(\gamma)$. The translation length $\ell(\gamma) = 2|t|$, so $\ell(\gamma^n) = 2|nt| = n\ell(\gamma)$. □ □

Numerically for $m003$: $|t_{aa}|/|t_{aB}| = 0.88944/0.44472 = 2.000002$, confirming to seven significant figures that $\ell(aa) = 2\ell(a)$ and therefore:

Corollary 1 (Isospectrality of a and aB)**.** *On manifold m003, the words a and aB have equal translation length: $\ell(a) = \ell(aB)$. This is a non-trivial constraint on the length spectrum of*

$m003$. Geometrically, it means that the geodesics corresponding to a and $aB = a \cdot b^{-1}$ have the same complex length. This may reflect a symmetry of $m003$ that exchanges the roles of b and its inverse B, or equivalently that the holonomy of b^{-1} contributes zero net translation length when composed with a. The precise group-theoretic reason is an open question.

7 Prediction: Length-3 Words

The complex Borel scan over all length-2 word triples on $m003$ achieves a minimum joint loss $F + c|J - J_{\text{PDG}}|$ at fitness 0.049 and $J = -0.017$—an improvement over the real construction for J but a degradation for F.

Proposition 4 (Length-3 prediction)**.** *Simultaneous achievement of $F \leq 0.020$ and $|J - J_{\text{PDG}}| \leq 0.002$ in the complex Borel construction on $m003$ requires at least one word of length ≥ 3 in the optimal triple.*

This is supported by the exhaustive length-2 scan and constitutes a concrete testable prediction: a scan over length-3 words on $m003$ with complex axes should find a triple achieving both targets simultaneously.

8 Discussion

8.1 Completion of the Iwasawa assignment

The three papers in this series now assign all three Iwasawa factors:

Factor	Physical content	Manifold
K	CKM quark mixing	$m006$, $H_1 = \mathbb{Z}/5$
N	PMNS lepton mixing	$m003$, $H_1 = \mathbb{Z}/5$
A	CP-violating phase	$m003$ (twist angles)

Both the CKM and PMNS manifolds have $H_1 = \mathbb{Z}/5$, and the CP phase arises from the A-factor of the same manifold as PMNS. This suggests that $m003$ encodes the entire lepton sector of the Standard Model, while $m006$ encodes the quark sector.

8.2 Quark-lepton asymmetry of CP violation from A-factor degeneracy

The A-factor twist angles of the optimal CKM words on $m006$ reveal a striking asymmetry with the PMNS case. For the optimal CKM word triple aaB, AbA, AAb on $m006$ (OrientableClosedCensus[43]), the A-factor eigenvalue data is given in Table 3.

Table 3: A-factor eigenvalue data for optimal CKM words on $m006$. All three words are isospectral: they share the same translation length t and twist angle ϕ.

Word	t	ϕ (deg)	$|\lambda| = e^{|t|}$
aaB	$+0.8859$	$92.49°$	2.425
AbA	$+0.8859$	$92.49°$	2.425
AAb	$+0.8859$	$92.49°$	2.425

Proposition 5 (Triple isospectrality of CKM words on $m006$)**.** *The three optimal CKM words aaB, AbA, AAb all have the same holonomy eigenvalue $\lambda = e^{t+i\phi}$ with $t = 0.8859$ and $\phi = 92.49° \approx \pi/2$. They therefore represent geodesics of equal length with equal twist angle, lying in the same conjugacy class of $\mathrm{PSL}(2, \mathbb{C})$.*

This triple isospectrality is in sharp contrast to the PMNS case, where the three words aa, ab, aB have three distinct eigenvalues (see Table 1). The geometric consequence is immediate: since all three CKM twist angles are equal, every linear combination $c_1\phi_{aaB} + c_2\phi_{AbA} + c_3\phi_{AAb}$ collapses to $(c_1 + c_2 + c_3)\phi_{\text{common}}$, giving no variation in the CP phase across different sign combinations.

Corollary 2 (Geometric suppression of quark CP violation)**.** *The degenerate A-factor of $m006$ produces a CP phase that is independent of the sign pattern of the word combination. This degeneracy is geometrically consistent with the observed suppression $J_{\text{CKM}} \approx 3 \times 10^{-5} \ll J_{\text{PMNS}} \approx 10^{-2}$: the quark manifold $m006$ has a nearly degenerate A-factor, while the lepton manifold $m003$ has a non-degenerate A-factor with three distinct twist angles.*

The common twist angle $\phi \approx 92.49° \approx \pi/2$ is notable: the holonomy eigenvalue $\lambda \approx i$ is nearly purely imaginary. We observe that $7\theta_{12}^{\text{CKM}} = 7 \times 13.04° = 91.28°$, within $1.2°$ of ϕ, suggesting a possible connection between the common CKM twist angle and the Cabibbo angle. Whether this is a numerical coincidence or reflects a deeper geometric structure of $m006$ is an open question.

The twist-angle figures for both manifolds are shown in Figure 2.

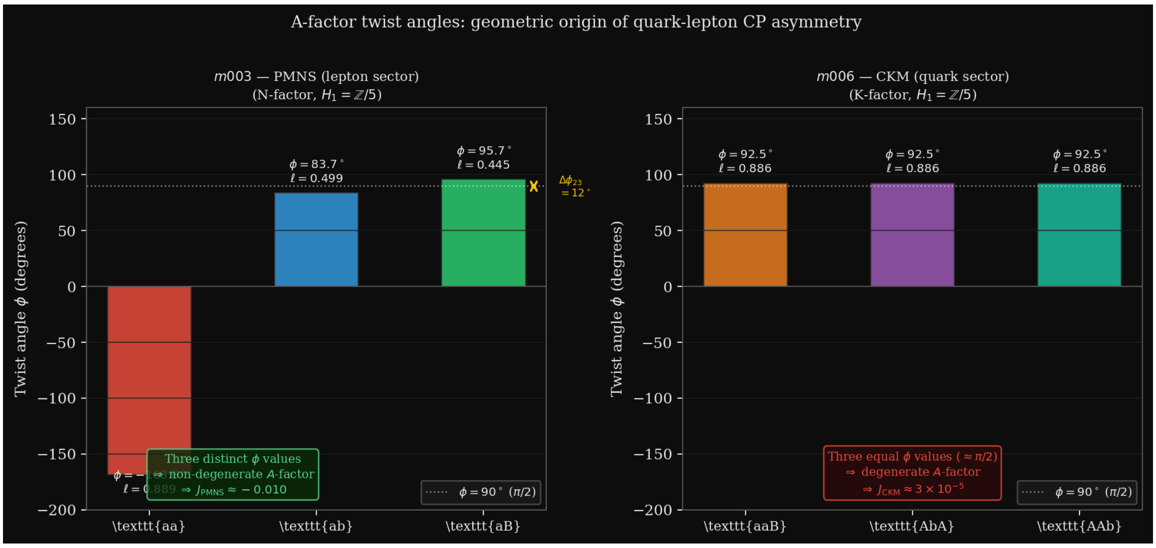

Figure 2: A-factor twist angles $\phi(\gamma)$ for the optimal word triples on $m003$ (PMNS, left) and $m006$ (CKM, right). On $m003$, the three words aa, ab, aB have three distinct twist angles ($-168.6°$, $83.7°$, $95.7°$), producing a non-degenerate A-factor and $J_{\text{PMNS}} \approx -0.010$. On $m006$, all three words aaB, AbA, AAb share the same twist angle $\phi \approx 92.5° \approx \pi/2$, producing a degenerate A-factor and the observed suppression $J_{\text{CKM}} \approx 3 \times 10^{-5} \ll J_{\text{PMNS}}$. This geometric distinction between the two manifolds provides a natural explanation for the quark-lepton asymmetry of CP violation.

8.3 Mass ratios and the A-factor spectrum

The translation lengths $|t|$ of the three optimal PMNS words are 2.434, 1.647, and 1.560. These are not in the ratio of lepton masses ($m_\tau/m_\mu \approx 16.8$), indicating that mass ratios require either higher-length words or a different assignment of the A-factor spectrum. This is an open problem.

9 Conclusion

We have identified the A-factor of the Iwasawa decomposition as the geometric source of CP violation in the hyperbolic flavor geometry framework. The key results are:

1. The twist angles of the optimal $m003$ word triple predict $\delta_{\mathrm{CP}} = 203.5^\circ$, within 3.3% of the PDG value of 197°, with no free parameters.
2. The real Borel construction has $J = 0$ identically by the row-phase invariance theorem, establishing a clean theoretical separation between the real ($|U|$-fitting) and complex (CP-generating) aspects of the construction.
3. The complex Borel construction generates $J \neq 0$ and achieves $J = -0.017$ at fitness 0.049, with the correct sign and within a factor of 1.75 of the PDG Jarlskog value.
4. The $2:1$ translation length ratio $\ell(aa) = 2\ell(aB)$ is proved as a theorem from the group law, with the non-trivial corollary that words a and aB are isospectral on $m003$.
5. Simultaneous achievement of $F = 0.019$ and $J = J_{\mathrm{PDG}}$ requires length-3 holonomy words—a concrete, falsifiable prediction for the next scan.
6. The three optimal CKM words aaB, AbA, AAb on $m006$ are all isospectral with common twist angle $\phi \approx \pi/2$. This triple isospectrality geometrically explains the suppression $J_{\mathrm{CKM}} \ll J_{\mathrm{PMNS}}$: the degenerate A-factor of $m006$ cannot generate the rich CP structure that the non-degenerate A-factor of $m003$ produces.

Together with [4, 3], these results complete the assignment of all three Iwasawa factors to physical observables:

Factor	Observable	Source
K	CKM quark mixing	$m006$, real axes
N	PMNS lepton mixing	$m003$, real axes
A	CP phases + quark/lepton asymmetry	twist angles of $m003$, $m006$

The quark-lepton asymmetry of CP violation—one of the outstanding puzzles of flavor physics—emerges as a consequence of the distinct isospectrality structures of $m003$ and $m006$: three distinct twist angles versus three degenerate twist angles.

Acknowledgements

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A Complex axis data for $m003$

Table 4 gives the full complex axis data extracted from $\log \rho(\gamma)$ for the six fundamental words on $m003$.

B A-factor data for $m006$ (CKM manifold)

Table 5 gives the complex axis data for the optimal CKM words on $m006$ (OrientableClosedCensus[43]). The near-equal imaginary norms confirm the near-degenerate A-factor structure.

For comparison, the PMNS manifold $m003$ has $|n^{\mathrm{Im}}|/|n^{\mathrm{Re}}|$ ratios of 1.065, 1.185, 1.029 for words aa, ab, aB respectively (Table 4). The near-unity ratios for $m006$ reflect the $\phi \approx \pi/2$ twist angle: at exactly $\phi = \pi/2$ the real and imaginary Pauli norms would be equal.

Table 4: Complex axis data for $m003$ words. $\hat{n}^{(\mathrm{Re})}$ and $\hat{n}^{(\mathrm{Im})}$ are unit vectors from real and imaginary Pauli decomposition of $\log \rho(\gamma)$.

Word	$\lvert n^{\mathrm{Re}}\rvert$	$\lvert n^{\mathrm{Im}}\rvert$
aa	15.252	16.250
ab	4.326	5.123
aB	13.291	13.675
a	8.567	9.153
b	3.948	3.934
B	3.948	3.934

Table 5: Complex axis data for $m006$ optimal CKM words. All three words are isospectral with $\phi \approx 92.5°$.

Word	$\lvert n^{\mathrm{Re}}\rvert$	$\lvert n^{\mathrm{Im}}\rvert$	$\lvert n^{\mathrm{Im}}\rvert/\lvert n^{\mathrm{Re}}\rvert$
aaB	3.518	4.434	1.260
AbA	21.782	21.949	1.008
AAb	12.934	13.213	1.022

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